

AIVerse 2.0 - Master Query for DeepSeek V4 / AI Coding Assistant

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You are an expert full-stack engineer. I need you to fix and complete a Next.js 14 AI Agent Marketplace application called AIVerse 2.0. Below is a comprehensive list of ALL bugs, errors, missing features, and required implementations. Fix them one by one, ensuring each change is production-ready.

The stack is: Next.js 14 + TypeScript + Tailwind CSS v4 + Prisma ORM + PostgreSQL 17 + Redis + NextAuth v4 + Stripe + BullMQ + Three.js + ioredis + Zod.

The Prisma schema already has these models: User, Account, Session, PasswordReset, EmailVerification, Referral, CreatorProfile, Agent, AgentVersion, AgentReview, AgentExecution, AgentFavorite, Wallet, Transaction, Subscription, Payment, CreatorPayout, File, AnalyticsEvent, SystemConfig, AuditLog.

DATABASE_SCHEMA_ATTACH: Read the file at prisma/schema.prisma in the project.

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SECTION 1: CRITICAL BUGS (Fix First)

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BUG 1.1: Stripe webhook handler has WRONG amountUsd field

File: src/app/api/webhooks/stripe/route.ts

Problem: Line 71 sets `amountUsd: credits` which stores credit count as USD amount. Should store actual dollar amount from checkout session.

Fix: Use the actual amount from the checkout session (checkoutSession.amount_total / 100), not the credit count.

BUG 1.2: Stripe webhook stores payment with wrong amountUsd type

File: src/app/api/webhooks/stripe/route.ts

Problem: Payment.amountUsd is Decimal in schema but code passes integer. Prisma may coerce incorrectly.

Fix: Pass as Decimal: `amountUsd: new Decimal(checkoutSession.amount\_total / 100)` or string.

BUG 1.3: Redis client returns null cast to Redis type - causes runtime errors

File: src/lib/redis.ts

Problem: Lines 11-12 return `null` as unknown as Redis` when REDIS\_URL is missing. This causes `redis.status` checks to throw "Cannot read property 'status' of null" in many places.

Fix: Make the redis export properly nullable and update ALL consumers (cacheGet, cacheSet, cacheDel, safety.ts, queue.ts, execute/route.ts) to handle null redis gracefully.

BUG 1.4: Agent creation allows duplicate slug with race condition

File: src/app/api/agents/route.ts

Problem: The try/catch P2002 retry uses Date.now() in slug, but if two requests come simultaneously, both may get same timestamp.

Fix: Use crypto.randomUUID() or nanoid(4) suffix instead of Date.now().

BUG 1.5: Agent execution does not increment totalRuns

File: src/app/api/agents/[slug]/execute/route.ts

Problem: After successful execution, totalRuns on the Agent model is never incremented.
 Fix: Add `await tx.agent.update({ where: { id: agent.id }, data: { totalRuns: { increment: 1 } } })`
 ↳ inside the credit deduction transaction.

BUG 1.6: Agent execution does not update avgRating
 File: src/app/api/agents/[slug]/execute/route.ts
 Problem: avgRating field on Agent is never updated when reviews are created.
 Fix: When a review is created in /api/agents/[slug]/reviews/route.ts, recalculate avgRating for the
 ↳ agent.

BUG 1.7: Agent execution credit deduction uses string comparison issue
 File: src/app/api/agents/[slug]/execute/route.ts
 Problem: The wallet.balance is Float in Prisma but creditsPerRun is Int. The comparison `wallet.balance
 ↳ < cost` could have floating point issues.
 Fix: Ensure consistent numeric types or use Prisma Decimal.

BUG 1.8: Session history loads wrong role format
 File: src/app/api/agents/[slug]/execute/route.ts
 Problem: loadHistory returns `{ role: string; content: string }[]` but appendHistory saves with roles
 ↳ "user"/"assistant". The streaming code maps these correctly BUT the type annotation says `role:
 ↳ string` which is loose.
 Fix: Use stricter types and ensure history mapping handles both "assistant" and "model" roles.

BUG 1.9: Trust score recordViolation has wrong score math
 File: src/lib/ai/safety.ts
 Problem: Line 141: `trust.score = Math.max(0, trust.score - 15 \* trust.violations)` — this subtracts 15
 ↳ * cumulative violations each time, causing exponential score drops. First violation: -15, second:
 ↳ -30, third: -45 = 10 points total.
 Fix: Should be `trust.score = Math.max(0, trust.score - 15)` — fixed penalty per violation.

BUG 1.10: OAuth sign-in does not create wallet for existing users
 File: src/lib/auth-config.ts
 Problem: Lines 81-105 create wallet only for NEW users (line 95-99). Existing users who sign in via
 ↳ OAuth for the first time don't get a wallet.
 Fix: Create wallet for ALL OAuth users on first sign-in, not just new accounts.

BUG 1.11: Creator earnings should not be added to "lifetimeEarned" for the user wallet
 File: src/app/api/agents/[slug]/execute/route.ts
 Problem: Line 106 in creditCreator uses `lifetimeEarned: { increment: share }` but lifetimeEarned on
 ↳ Wallet tracks purchased credits, not creator earnings. Creator earnings go to a separate
 ↳ CreatorProfile.totalEarned.
 Fix: Update CreatorProfile.totalEarned instead of Wallet.lifetimeEarned for creator earnings.

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SECTION 2: MISSING MVP FEATURES (Implement)

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FEATURE 2.1: Implement pgvector semantic search for agents
 File: src/app/api/agents/route.ts
 Currently the search uses basic `contains` (LIKE queries). The schema already has `embedding
 ↳ Unsupported("vector(1536)")` field.

Implement: When search query is provided, first generate embedding via OpenRouter, then use `prisma.
↪ \$queryRaw` with pgvector `<=>` cosine distance operator to rank results. Fall back to text search if
↪ embedding generation fails.

FEATURE 2.2: Implement proper trending algorithm

File: src/app/api/agents/route.ts

Currently sorts by `totalRuns` (all-time). Per FRD, trending should be based on executions in last 7
↪ days.

Implement: Add a `trending` sort option that counts agent_executions WHERE createdAt >= NOW() - INTERVAL
↪ '7 days', or maintain a rolling counter.

FEATURE 2.3: Multi-step agent publishing wizard

File: src/app/agents/create/page.tsx

Currently a single form. Per FRD, should be 5-step wizard:

Step 1: Name, category, description, tags (auto-generate slug)

Step 2: System prompt (rich text editor, max 8000 chars, character counter)

Step 3: Tools config (MCP tool selection - show disabled/state for MVP)

Step 4: Pricing (1-100 credits per run)

Step 5: Test in sandbox (max 5 test runs), then submit for review

FEATURE 2.4: Sandbox test mode enforcement

File: src/app/api/agents/[slug]/execute/route.ts

The route checks `body.\_test === true` for test mode but there's no enforcement of max 5 test runs per
↪ agent.

Implement: Track test run count per agent per creator in Redis or database. Block after 5 test runs.

FEATURE 2.5: Auto-generate slug from name

File: src/lib/utils.ts and src/app/agents/create/page.tsx

Remove manual slug field from form. Auto-generate from name using slugify() with random 4-char suffix on
↪ collision (per FRD).

FEATURE 2.6: Admin user management - search, suspend, credit adjustment

File: src/app/admin/page.tsx

The admin page shows users but has no actions. Add:

- Search users by email/username
- Suspend/unsuspend user (set isActive=false, invalidate sessions)
- Manual credit adjustment (post to /api/admin/users/[id]/credits)

FEATURE 2.7: System prompt encryption at rest

File: src/lib/encryption.ts (create) and src/app/api/agents/route.ts

Per TRD, system prompts should be AES-256 encrypted. Create encryption utility using Node.js crypto
↪ module with ENCRYPTION_KEY from env. Encrypt on save, decrypt on read (admin moderation and
↪ execution).

FEATURE 2.8: Email verification required before agent publish

File: src/app/api/agents/route.ts

Line 17-20 checks isActive but OAuth users may skip verification. Ensure all non-verified users are
↪ blocked from publishing.

FEATURE 2.9: Password reset token expiry not enforced

File: src/app/api/auth/reset-password/route.ts

The token expiry (1 hour) may not be checked. Ensure the route checks `expiresAt > now()` and marks token
↪ as used via `usedAt`.

FEATURE 2.10: Email verification token expiry not enforced

File: `src/app/api/auth/verify-email/route.ts`

Ensure token expiry is checked and tokens are single-use.

FEATURE 2.11: Creator payout system completion

File: `src/app/api/payouts/request/route.ts`

Ensure: minimum \$10 threshold, 30-day hold on first payout, creator earnings tracking via

↪ `CreatorProfile`.

FEATURE 2.12: Agent version control on update

File: `src/app/api/agents/[slug]/route.ts` (PUT)

When an agent is updated, create a new `AgentVersion` record with the old config snapshot. Allow viewing

↪ version history.

FEATURE 2.13: PII redaction in execution logs

File: `src/app/api/agents/[slug]/execute/route.ts`

Use the existing `sanitizeInput` function from `safety.ts` on logged messages. Ensure `errorLog` field also

↪ gets redacted.

FEATURE 2.14: GDPR data export endpoint

File: `src/app/api/users/me/export/route.ts`

Implement full data export: user profile, agents, executions, transactions, reviews, wallet, payouts,

↪ all in JSON format downloadable.

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SECTION 3: CODE QUALITY & PERFORMANCE FIXES

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FIX 3.1: Add pagination to all list endpoints

Files: `src/app/api/agents/route.ts`, `src/app/api/wallet/transactions/route.ts`, `src/app/api/admin/users/`

↪ `route.ts`

Ensure all list endpoints support cursor-based pagination with `limit` parameter.

FIX 3.2: Add proper error handling middleware

Create: `src/lib/error-handler.ts`

Standardize all API error responses to match TRD format: `{ error: string, code: string, details?:`

↪ `object }`

FIX 3.3: Add request timeout and abort signal

Files: All API routes

Add `AbortSignal.timeout(10000)` or middleware for route timeouts.

FIX 3.4: Add request logging

File: `src/lib/logger.ts`

Ensure all API requests are logged with `method`, `path`, `userId`, `duration`.

FIX 3.5: Add rate limiting to all sensitive endpoints

File: `src/lib/rate-limit.ts`

Ensure rate limiting on: register, login, forgot-password, execute, checkout, payouts.

FIX 3.6: Add database connection pooling

File: src/lib/prisma/index.ts

Add Prisma connection pool configuration for production.

FIX 3.7: Fix Three.js memory leak

File: src/components/effects/neural-background.tsx

The useEffect cleanup disposes renderer but doesn't dispose all geometries/materials properly. Ensure
↪ complete cleanup.

FIX 3.8: Add loading skeletons to pages

Files: All page loading.tsx files

Replace basic loading spinners with proper skeleton loaders matching page layout.

FIX 3.9: Add OpenAPI/Swagger documentation

Create: src/app/api-docs/page.tsx

Generate API documentation from route handlers.

FIX 3.10: Add proper TypeScript types for all `any` usages

Files: Multiple files

Replace all `any` types with proper interfaces. Priority: API routes, gateway, auth.

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SECTION 4: SECURITY HARDENING

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SEC 4.1: Add CSRF protection for state-changing operations

Implement: Double-submit cookie pattern or CSRF tokens for POST/PUT/DELETE.

SEC 4.2: Add Content Security Policy headers

File: next.config.mjs

Add CSP headers to prevent XSS. Allow 'self', Stripe, Google OAuth, Cloudflare Turnstile domains.

SEC 4.3: Add security headers

File: next.config.mjs

Add: X-Frame-Options, X-Content-Type-Options, Referrer-Policy, Permissions-Policy.

SEC 4.4: Input validation on file uploads

File: src/app/api/upload/route.ts

Validate file types, sizes, scan for malware. Limit to 10MB free / 50MB Pro.

SEC 4.5: SQL injection prevention audit

Files: All raw query locations

Ensure `\$queryRaw` uses parameterized queries only. Check src/app/api/agents/route.ts line 103.

SEC 4.6: Add API key rotation support

File: src/lib/ai/routing.ts

Support multiple API keys per provider for rotation on rate limits.

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SECTION 5: ENVIRONMENT VARIABLES VALIDATION

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Create: src/lib/env.ts

Implement strict env validation using Zod at startup. Required vars:

- DATABASE_URL (required)
- AUTH_SECRET (required, min 32 chars)
- REDIS_URL (optional in dev, required in prod)
- At least one AI provider API key
- STRIPE_SECRET + STRIPE_WEBHOOK_SECRET (if payments enabled)
- RESEND_API_KEY (if email enabled)

Fail fast with clear error message if required vars missing.

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SECTION 6: IMPLEMENTATION PRIORITY

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Priority 1 (Launch Blockers):

- BUG 1.1, 1.3, 1.5, 1.9, 1.10
- FEATURE 2.1 (semantic search fallback)
- FEATURE 2.5 (auto-slug)
- SEC 4.2, 4.3 (security headers)

Priority 2 (Pre-Launch):

- BUG 1.2, 1.4, 1.6, 1.11
- FEATURE 2.2, 2.3, 2.4
- FIX 3.1, 3.2, 3.5

Priority 3 (Post-Launch):

- FEATURE 2.6 through 2.14
- FIX 3.3, 3.4, 3.6 through 3.10
- SEC 4.1, 4.4, 4.5, 4.6

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IMPORTANT NOTES:

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1. Do NOT upgrade to Next.js 15 or NextAuth v5 - stick with current versions.
2. Do NOT add Vercel AI SDK - the custom provider system works well.
3. Keep all existing UI components and styling intact.
4. Write tests for all new features (Vitest).
5. Ensure all changes are TypeScript strict-compliant.
6. Run `npm run typecheck` and `npm run lint` after all changes.
7. Maintain the glassmorphism UI design system.

How to Use This Query

1. **Open your AI coding tool** (DeepSeek V4, Cursor, etc.)
2. **Load the entire project** as context

3. **Paste the above prompt** (everything between the triple backticks)
4. Let the AI process and implement fixes section by section
5. Review each change before applying
6. Run `npm run typecheck` after each batch of changes

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